Running it in production

Deploy it, watch it, know what it costs, and be able to roll it back at 2 am.

60 min | Recorded | Class 14 of 15

14

What this session covers

01
Shipping it
2 slides

02
Keeping it right
2 slides

03
Changing it safely
2 slides

04
Code walkthrough
4 slides

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Section 1 of 4

Shipping it

A notebook that worked once isn't a service.

01

Getting from a script to a service takes the same five things every time

02

Every call logs what it cost, how long it took, and which prompt it used

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Getting from a script to a service takes the same five things every time

1

Package it in a container

2

Keys from a secret store, not the code

3

Trace every call

4

Log cost and latency per call

5

A way to roll back in one step

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Every call logs what it cost, how long it took, and which prompt it used

Prompt version, model, tokens in and out, latency, cost. One line per call.

Without this, the first you hear of a problem is the invoice from the provider, or a user.

latency budget per ticket	2 s
cost budget per ticket	$0.002
error budget	1%

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Budgets are examples. Set real ones per feature and alert when they’re crossed.

Section 2 of 4

Keeping it right

Things change underneath it.

01

Version everything that can change the answer

02

Drift is when the world changes and the model doesn’t

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Version everything that can change the answer

The model	Providers update models behind the same name. Pin the version.
The prompt	A one-word change can move the score. Give prompts version numbers, like code.
The data	Runbooks get rewritten; the retrieval index has to follow.
The test set	Frozen, but grown with every production failure (class 13).

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Drift is when the world changes and the model doesn’t

New supplier
A new cleaning contractor replaces D.I.H.. Tickets say a name the model never saw.

New process
EU HS codes change in July. Last month’s right answer is this month’s wrong one.

New model
The provider upgrades. Same prompt, different answers.

Rerun the test set on a schedule and on every change. A drop is the only early warning you get.

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Section 3 of 4

Changing it safely

When to ship, and how to take it back.

01

No changes to anything finance touches during period close

02

Coding assistants make more code; they don't make more reviewers

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No changes to anything finance touches during period close

Ledgerline, PayGrid and InvoiceHub are busiest during the last two days of the month and the first three of the next. That's when a broken change costs most and is hardest to spot.

Ship the week after close. Roll back first and investigate second.

change freeze per month 5 days

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Coding assistants make more code; they don’t make more reviewers

What gets faster

- Writing the first version
- Tests, boilerplate, glue
- Trying three approaches instead of one

What doesn’t

- Reading it properly
- Knowing it’s right for our data rules
- Being on call for it at 2 am

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Section 4 of 4

Code walkthrough

A container, a trace viewer and a cost log.

01
Compose runs the service and a trace viewer side by side

02
The cost log is one JSON line per call

03
Where today sits on the card

04
What to do next

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Compose runs the service and a trace viewer side by side

code/class14/compose.yaml

```yaml
services:
  triage:
    build: .
    environment:
      LLM_API_KEY: ${LLM_API_KEY}
      LLM_MODEL: ${LLM_MODEL}
      PROMPT_VERSION: "triage-v3"
      OTEL_EXPORTER_OTLP_ENDPOINT: http://jaeger:4318
  jaeger:
    image: jaegertracing/all-in-one:1.60
```


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The cost log is one JSON line per call

code/class14/costlog.py

```python
record = {
    "prompt_version": prompt_version, "model": reply.model,
    "tokens_in": u.prompt_tokens, "tokens_out": u.completion_tokens,
    "latency_ms": round((time.perf_counter() - start) * 1000),
    "cost_usd": round((u.prompt_tokens * PRICE["input"] + u.completion_tokens *
PRICE["output"]) / 1e6, 6),
}
```


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Where today sits on the card

1. A script	The steps are known and don't change.
2. A rule	The decision is known and must be explainable line by line.
3. Classical ML	**Labelled history, and the input is a table or short text.**
4. A small trained model	**Lots of text, images or audio, enough labels, cost per call matters.**
5. An LLM with a prompt	**Varied language in and out, few labels, someone checks the output.**
6. An LLM plus retrieval, tools or agents	**Needs private knowledge, has to act, or takes several steps.**
7. Not yet	No ground truth, needs a guarantee, or nobody owns the result.

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What to do next

1. Run `docker compose up` in `code/class14` and find one trace.
2. Change the prompt version, rerun the class 13 test set, and compare.
3. Pick a problem for the capstone: one of yours, or draw one on the day.

Questions after class: the course channel.

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